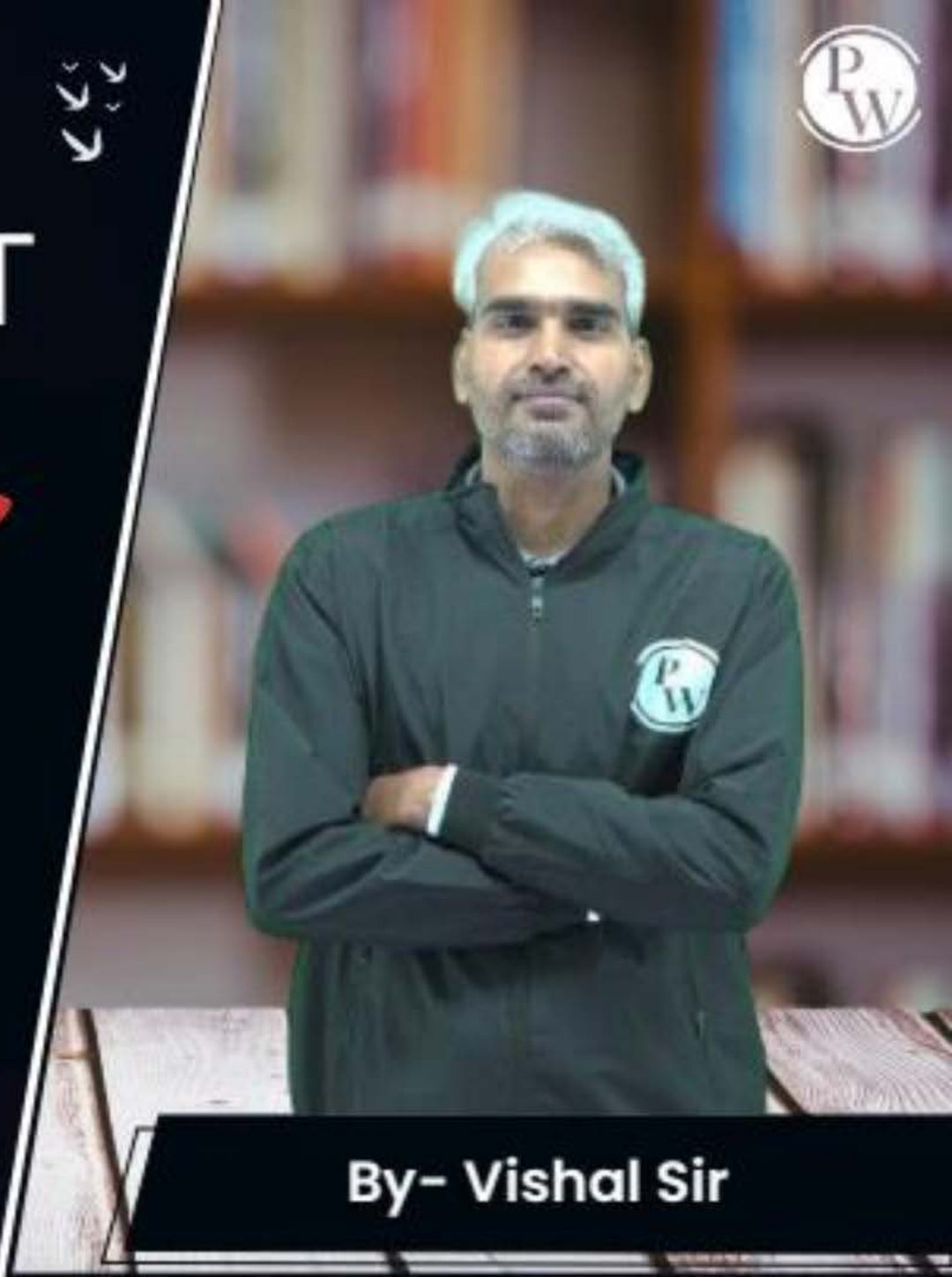
Computer Science & IT



Discrete Mathematics

Mathematical Logic

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Euler train, Euler circuit, and traversable graph
- **Topic** Hamiltonian path and Hamiltonian cycle
- **Topic** Distance, Eccentricity, Diameter, Radius and Girth
- **Topic** n-cube graph
- **Topic** Clique

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Topics to be Covered



Topic — Propositions and their types

Topic — Connectives

Topic — Tautology, Contradiction, Contingency and Satisfiable propositional functions

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Mathematical Logic

1. Propositional Logic
2. Predicate Logic



Topic : Propositional Logic

Proposition :—

A declarative sentence to which we can assign only one of the truth values (i.e. TRUE/FALSE) is called a propositional statement or proposition.

eg: ① It rains today

② Cricket match is played today



Topic : Assumptions of proposition

- Law of excluded middle : If a proposition is not true then it will be false, and similarly if a proposition is not false then it will be true.

- Law of contradiction : A proposition can not be true as well as false simultaneously.



Topic : Atomic propositions

- Atomic propositions are simple propositions which can not be divided further.

Eg: ① It rains today ⟶ "p"

② Cricket match is played ⟶ "q"



Topic : Compound Propositions

- Compound propositions are constructed by combining atomic propositions with the help of connectives.

- Compound propositions are also known as propositional formula, or propositional function.



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eg: It rains today (p) Or (∨ Disjunction) Cricket match is played (q) = f(p, q) = p ∨ q



Topic : Compound Propositions

- Compound propositions are constructed by combining atomic propositions with the help of connectives.

Compound propositions are also known as propositional formula, or propositional function.

eg: It rains today OR Cricket match is played

It rains today → 1st atomic Proposition

OR → Connective

Cricket match is played → 2nd atomic Proposition

Compound Proposition



Topic : Connectives

1. Negation (¬/~) / Not
2. AND (∧) / Conjunction
3. OR (∨) / Disjunction
4. Implication (→)
5. Biconditional (↔)

Most important



Topic : Negation (¬/~)

- If p is any proposition, then negation of p / not(p) / ¬p is also a proposition whose truth value is true only when p is false and it is false only when p is true.

p	~p
T	F
F	T



Topic : AND (∧) / Conjunction

If p and q are any two propositions, then “p AND q” / p ∧ q is also a proposition whose truth value is true only when both p as well as q are true.

p	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F



Topic : OR (V) / Disjunction

If p and q are any two propositions, then “p OR q” / p V q is also a proposition whose truth value is false only when both p as well as q are false.

p	q	p V q
T	T	T
T	F	T
F	T	T
F	F	F



Topic : Implication (→)

↑ aka (if-then)

If p and q are any two propositions, then p implies q (i.e. p → q / if p then q) is also a proposition whose truth value is false only when p is true, and q is false.

p → q

antecedent — Consequent

p	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

If it rains today then Cricket match will not be played $\equiv p \rightarrow q$

(p: it rains today; q: Cricket match will not be played)

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- $p \to q$ and $\sim p \vee q$ have Same truth tables
∴ $p \to q \equiv \sim p \vee q$

Different truth tables
∴ $p \to q \not\equiv q \to p$

$q \to p \cong \sim p \to \sim q$

p	q	$\sim p$	$\sim q$	$p \to q$	$\sim p \vee q$	$q \to p$	$\sim p \to \sim q$	$\sim q \to \sim p$
T	T	F	F	T	T	T	T	T
T	F	F	T	F	F	T	T	F
F	T	T	F	T	T	F	F	T
F	F	T	T	T	T	T	T	T

$p \to q \not\equiv \sim p \to \sim q$

$p \to q \equiv \sim q \to \sim p$



Topic : Implication (→)

- If p is true, then q must be true for p→q to be true.
- If p is false, then p→q is always true whatever is the truth value of q.
- If q is true, the p→q is always true whatever is the truth value of p.
- Truth table of p→q and truth table of ¬p V q are exactly same, therefore both are equivalent.
- Converse of p→q is q→p.
- Opposite or Inverse of p→q is "¬p → ¬q"
- Contra positive for p→q is "¬q → ¬p"

$$p \rightarrow q \equiv \sim p \vee q$$

$$p \rightarrow q \equiv \underbrace{\sim q \rightarrow \sim p}_{\text{Contra positive of } p \rightarrow q}$$

$$p \rightarrow q \not\equiv \sim p \rightarrow \sim q$$
(inverse of $p \rightarrow q$)

$$p \rightarrow q \not\equiv q \rightarrow p$$
(converse of $p \rightarrow q$)

$$q \rightarrow p \equiv \sim p \rightarrow \sim q$$

Converse of $p \rightarrow q$ is equivalent to inverse of $p \rightarrow q$



Topic : Biconditional (↔)

(if and only if)

If p and q are any two propositions, then "p ↔ q" (i.e. p *if and only if* q) is also a proposition whose truth value is true only when both p and q have same truth value.

Topic : Biconditional (↔)

(if and only if)

- If p and q are any two propositions, then "p ↔ q" (i.e. p *if and only if* q) is also a proposition whose truth value is true only when both p and q have same truth value.

$f_1(p,q)$ $f_2(p,q)$

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$



Topic : Biconditional (↔)

- We observed $(p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

"$p \leftrightarrow q$" is true if and only if both "$p \rightarrow q$" and "$q \rightarrow p$" are true



Topic : Tautology

A propositional function which is always true is called tautology.

A propositional function which is always true is also called a valid propositional function.



Topic : Tautology

{ Tautology is term which is defined w.r.t. propositional logic }

- A propositional function which is always true is called tautology.

A propositional function which is always true is also called a valid propositional function.

W.r.t. propositional logic, Tautologies & Valid propositional function are same things

{ But it will not be the case w.r.t. Predicate logic }

eg: $p \vee \sim p$ — $f_1(p)$

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T

Always true
∴ Tautology

eg: Check if $p \rightarrow (q \rightarrow p)$ is a tautology or not — $f_2(p,q)$

p	q	$(q \rightarrow p)$	$p \rightarrow (q \rightarrow p)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	T

Always true
∴ $p \rightarrow (q \rightarrow p)$ is a tautology
(or)
$p \rightarrow (q \rightarrow p)$ is a valid Propositional formula



Topic : Contradiction

- A propositional function which is always false is called Contradiction.

$f(p)$

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

Always false
∴ Contradiction

Note:- ① If a propositional function is always true then it is called Valid propositional function

② If propositional function is false for atleast one case, then it is an invalid Propositional function

- Each Contradiction is invalid propositional function, but Each invalid propositional function need not be Contradiction



Topic : Contingency

A propositional function which is neither a tautology nor a contradiction is called a contingency.

i.e., A contingency is true for at least one case as well as false for at least one case.

eg:

p	q	p→q
T	T	T
T	F	F
F	T	T
F	F	T

f(p,q)

True for at least one Combination as well as false for at least one Combination

∴ it is a Contingency.

Note: Every Contingency is invalid



Topic : Satisfiable

A propositional function which is true for at least one case is called a satisfiable propositional function.

- i.e. if a propositional function is not satisfiable, then it is a Contradiction



Topic : NOTE

- Every propositional function which is not a contradiction is satisfiable.

- Every tautology is satisfiable but not vice-versa.

  i.e. Every Satisfiable Propositional function need not be a tautology

- Every contingency is satisfiable, but every satisfiable function need not be contingency.

  eg: Tautology :⟶ Satisfiable, but not a Contingency



Topic : NOTE

Tautology	Contradiction	Contingency
Always True	Always False	Neither always true nor always false
Valid	Invalid	Invalid
Satisfiable	Not satisfiable	Satisfiable



2 mins Summary

Topic Propositions and their types

Topic Connectives

Topic Tautology, Contradiction, Contingency and Satisfiable propositional functions


THANK - YOU
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